HC-SR04 / HC-SR04P Ultrasonic Distance Sensor

Mechanical Dimensional Guide, Sonar Timing, & RP2040 Integration

2 cm – 400 cm | 40 kHz SONAR | 3.3V / 5.0V

1. OPERATIONAL OVERVIEW & SPECIFICATIONS

The HC-SR04 provides non-contact distance measurement using 40 kHz ultrasonic sound bursts. The module emits an 8-cycle sonic burst upon receiving a 10μs trigger pulse, then outputs an Echo pulse whose duration is proportional to round-trip transit time.

Parameter	Specification
Operating Voltage	5V DC (Classic HC-SR04) / 3.0V–5.5V (HC-SR04P)
Quiescent Current	< 2 mA
Working Current	15 mA
Effective Angle	< 15° measuring cone
Ranging Distance	2 cm – 400 cm (0.8" – 157")
Resolution / Accuracy	0.3 cm
Trigger Pulse Width	10 μs TTL pulse
Echo Pulse Output	TTL pulse width: $T = \frac{ ext{Distance (cm)} imes 58}{ ext{or Distance (in)} imes 148}$

2. PINOUT & INTERCONNECT TABLE

Pin	Name	Type	Pico Connection & Electrical Notes
1	`VCC`	Power	Connect to `VBUS` (5V) or `3V3` (HC-SR04P)
2	`Trig`	Input	Connect to Pico `GP5` (Trigger pulse)
3	`Echo`	Output	Connect to Pico `GP6` *(See 5V Protection Note)*
4	`GND`	Power	System Ground (Connect to Pico `GND`)

Critical 3.3V Logic Protection (Classic HC-SR04): If powered from 5V (VBUS), the Echo pin outputs a 5V signal. To safely connect to the RP2040's 3.3V-tolerant GPIOs, either use a **1kΩ / 2kΩ voltage divider** on the Echo line, or use the 3.3V-native **HC-SR04P** / **RCWL-1601** variant.

3. MECHANICAL DIMENSIONS & ENCLOSURE GEOMETRY

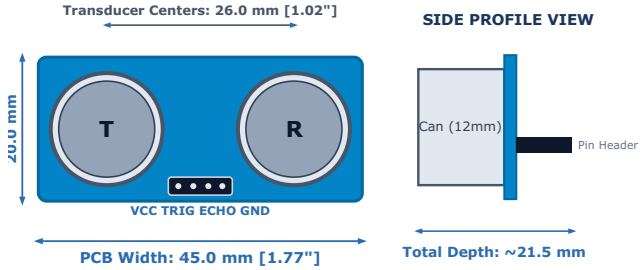


Figure 1: HC-SR04 Sensor Geometry, Dual Barrel Insets, and Enclosure Cutout Specs.

4. ENCLOSURE BEZEL DESIGN NOTES

Dual Barrel Cutouts: Drill two circular holes of **Ø 16.5 mm** spaced exactly **26.0 mm center-to-center**. Bevel or chamfer the front aperture edges to prevent acoustic reflections from triggering false near-field echos.

5. CIRCUITPYTHON LIBRARY SETUP

Install the hardware driver to your Pico's `lib/` folder:

- `adafruit_hcsr04.mpy` (Available via CircuitPython Bundle)

6. CIRCUITPYTHON READING LOOP

```
import time, board
import adafruit_hcsr04

sonar = adafruit_hcsr04.HCSR04(trigger_pin=board.GP5, echo_pin=board.GP6)

while True:
    try:
        dist = sonar.distance
        print(f"Distance: {dist:.1f} cm ({dist / 2.54:.1f} in)")
    except RuntimeError:
        pass  # Sensor timed out or echo missed
    time.sleep(0.05)
```